Isaac Dreves

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EDUCATION

Case Western Reserve University, Cleveland, OH — **Expected May 2028**

Bachelor of Science in Mechanical and Aerospace Engineering — Cumulative GPA: 3.76/4.0

Relevant Coursework: Thermodynamics, Statics, Dynamics, Differential Equations, Materials & Manufacturing Processes, Numerical Methods, Circuits & Instrumentation.

SKILLS

Software & Analysis: SolidWorks (CSWA, PDM, FEA), Fusion 360, MATLAB, Cura, Motion Studies

Hardware & Prototyping: Manual Mill & Lathe, Additive Manufacturing, Waterjet, Laser Cutting, Microcontrollers

Applied Engineering: GD&T, Design for Manufacturing, Material Selection, BOM Management, Technical Writing

PROJECTS & EXPERIENCE

VTOL Drone CWRU, Vertical Take-Off and Landing Club — **September 2024 - Present**

- Collaborating with a 50+ person multidisciplinary team to create autonomous and nonautonomous drones for the Design Build Vertical Flight Competition.
- Integrating Pugh Decision Matrices to evaluate design and manufacturing tradeoffs for efficiency, reliability, and budget constraints.
- Designed and prototyped a retractable landing gear system to improve aerodynamics and battery life; applied FEA to optimize strength-to-weight ratio, sourced components, and executed iterative CAD design and manufacturing within a small sub team.

Manual Trash Compactor, Consumer Product Design — **September 2025 - November 2025**

- Led a 4-person team to develop a manual trash compactor, focusing on Design for Manufacturing and cost effectiveness for commercial production.
- Performed compression force calculations and applied proper fits, tolerances, weld symbols, and sheet metal design principles to all parts.
- Produced 25+ fully dimensioned engineering drawings including assemblies, BOMs, and detailed part drawings. Conducted motion studies to validate design functionality, and created final commercial renders.

Sports Cone Material Selection, Material Science Research — **April 2025 – May 2025**

- Evaluated and selected Solid Silicone Rubber HCR to replace traditional LDPE, prioritizing improved flexibility, high friction coefficients, and UV resistance.
- Conducted trade off analyses comparing mechanical failure modes (cracking, fading) and long-term cost-effectiveness against existing industry standards.
- Outlined a rapid prototyping and manufacturing workflow using 3D printed molds, liquid resin casting, and iterative physical testing before mass production.

Hydroponics System, Independent Project — **July 2025 - Present**

- Designed, built, and tested a hydroponic system integrating pumps and LED lighting for continuous indoor crop production; achieving approximately 2x faster growth cycles and 4x yield per square foot for lettuce compared to traditional soil cultivation.
- Currently developing an Arduino based nutrient dosing system using pH and EC sensors for automated pump control.

MATLAB, Numerical Methods — **October 2025**

- Engineered a MATLAB algorithm for Newton's Divided Difference polynomial interpolation; utilized pointers, dynamic 2D arrays, and iterative loops to calculate approximations and achieve user-controlled convergence criteria without relying on built in MATLAB sorting functions.